CENTRAL BOARD OF SECONDARY EDUCATION

SAMPLE QUESTION PAPER

Academic Session: 2026–27

Class: 10

Subject: Mathematics

Maximum Marks: 80

Time Allowed: 3 hours

GENERAL INSTRUCTIONS

• This Question Paper has 5 Sections A, B, C, D, and E.

• Section A has 20 Multiple Choice Questions (MCQs) carrying 1 mark each.

• Section B has 5 Very Short Answer (VSA) type questions carrying 2 marks each.

• Section C has 6 Short Answer (SA) type questions carrying 3 marks each.

• Section D has 4 Long Answer (LA) type questions carrying 5 marks each.

• Section E has 3 Case Based integrated units of assessment (4 marks each) with sub-parts of the values of 1, 1, and 2 marks each respectively.

• All questions are compulsory. However, an internal choice in 2 questions of 2 marks, 2 questions of 3 marks, and 2 questions of 5 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E.

• Draw neat figures wherever required. Take $\pi = 22/7$ wherever required unless stated otherwise.

SECTION A

Multiple Choice Questions

Q1. If the sum of the zeroes of the quadratic polynomial $P(x) = kx^2 + 2x + 3k$ is equal to their product, then the value of k is:

(A) -2/3

(B) 2/3

(C) 1/3

(D) -1/3

(1 Mark)

Q2. The pair of equations $y = 0$ and $y = -7$ has:

(A) one solution

(B) two solutions

(C) infinitely many solutions

(D) no solution

(1 Mark)

Q3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is:

(A) 1

(B) 2

(C) 3

(D) 4

(1 Mark)

Q4. For what value of p are 2p + 1, 13, 5p - 3 three consecutive terms of an Arithmetic Progression?

(A) 4

(B) 5

(C) 6

(D) 7

(1 Mark)

Q5. In ΔABC, D and E are points on sides AB and AC respectively such that DE || BC. If AD/DB = 3/4 and AC = 14 cm, then AE is equal to:

(A) 6 cm

(B) 7 cm

(C) 8 cm

(D) 9 cm

(1 Mark)

Q6. The distance of the point P(-6, 8) from the origin is:

(A) 8 units

(B) $2\sqrt{7}$ units

(C) 10 units

(D) 6 units

(1 Mark)

Q7. If tan θ = 3/4, then the value of (1 - cos θ) / (1 + cos θ) is:

(A) 1/9

(B) 2/9

(C) 1/3

(D) 2/3

(1 Mark)

Q8. If the perimeter of a circle is equal to that of a square, then the ratio of their areas is:

(A) 22:7

(B) 14:11

(C) 7:22

(D) 11:14

(1 Mark)

Q9. The probability of an event E is 0.05. What is the probability of 'not E'?

(A) 0.95

(B) 0.05

(C) 0.5

(D) 1

(1 Mark)

Q10. The sum of the first 100 positive integers is:

(A) 5000

(B) 5050

(C) 5100

(D) 5150

(1 Mark)

Q11. A quadratic equation $ax^2 + bx + c = 0$ has distinct real roots if:

(A) $b^2 - 4ac > 0$

(B) $b^2 - 4ac < 0$

(C) $b^2 - 4ac = 0$

(D) $b^2 - 4ac \geq 0$

(1 Mark)

Q12. If the points (0, 0), (1, 0) and (0, 1) are the vertices of a triangle, then its area is:

(A) 1 $unit^2$

(B) 1/2 unit^2

(C) 2 unit^2

(D) 0 unit^2

(1 Mark)

Q13. The value of (sin 30° + cos 30°) - (sin 60° + cos 60°) is:

(A) -1

(B) 0

(C) 1

(D) 2

(1 Mark)

Q14. The ratio of the volume of a cone to that of a cylinder of the same base radius and same height is:

(A) 1:1

(B) 1:2

(C) 1:3

(D) 2:3

(1 Mark)

Q15. For the following distribution:

Class: 0-5, 5-10, 10-15, 15-20, 20-25

Frequency: 10, 15, 12, 20, 9

The upper limit of the modal class is:

(A) 10

(B) 15

(C) 20

(D) 25

(1 Mark)

Q16. If a circle has its center at (2, 3) and passes through the point (5, 7), then its radius is:

(A) 3 units

(B) 4 units

(C) 5 units

(D) 6 units

(1 Mark)

Q17. A tangent to a circle intersects it in:

(A) one point

(B) two points

(C) three points

(D) no point

(1 Mark)

Q18. The value of (1 + tan^2 A) / (1 + cot^2 A) is:

(A) sec^2 A

(B) -1

(C) cot^2 A

(D) tan^2 A

(1 Mark)

Q19. Assertion (A): The product of two consecutive positive integers is divisible by 2.

Reason (R): The product of two consecutive integers is always an even number.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

(1 Mark)

Q20. Assertion (A): The value of sin A is always less than 1.

Reason (R): The hypotenuse is the longest side in a right-angled triangle.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

(1 Mark)

SECTION B

Very Short Answer Questions

Q21. Find the value of k for which the system of equations $kx + 3y = k - 3$ and $12x + ky = k$ has infinitely many solutions.

(2 Marks)

Q22. In the given figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then find $\angle PTQ$.

(2 Marks)

OR

If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then find $\angle POA$.

(2 Marks)

Q23. If the sum of the first n terms of an AP is given by $S_n = 3n^2 + 5n$, find the common difference of the AP.

(2 Marks)

Q24. A die is thrown once. Find the probability of getting:

(i) a prime number

(ii) a number lying between 2 and 6

(2 Marks)

Q25. Prove that $\sqrt{3}$ is an irrational number.

(2 Marks)

OR

Find the HCF and LCM of 12, 15, and 21 by the prime factorization method.

(2 Marks)

SECTION C

Short Answer Questions

Q26. Prove that $(\sin \theta - 2\sin^3 \theta) / (2\cos^3 \theta - \cos \theta) = \tan \theta$.

(3 Marks)

Q27. Find the coordinates of the points of trisection of the line segment joining (4, -1) and (-2, -3).

(3 Marks)

Q28. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

(3 Marks)

OR

A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding major sector. (Use π = 3.14)

(3 Marks)

Q29. Prove that the lengths of tangents drawn from an external point to a circle are equal.

(3 Marks)

Q30. A fraction becomes 1/3 when 1 is subtracted from the numerator and it becomes 1/4 when 8 is added to its denominator. Find the fraction.

(3 Marks)

OR

The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

(3 Marks)

Q31. The median of the following data is 525. Find the values of x and y, if the total frequency is 100.

Class Interval: 0-100, 100-200, 200-300, 300-400, 400-500, 500-600, 600-700, 700-800, 800-900, 900-1000

Frequency: 2, 5, x, 12, 17, 20, y, 9, 7, 4

(3 Marks)

SECTION D

Long Answer Questions

Q32. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

(5 Marks)

OR

A motor boat whose speed in still water is 18 km/h, takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

(5 Marks)

Q33. State and prove Basic Proportionality Theorem (Thales Theorem).

(5 Marks)

Q34. A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π.

(5 Marks)

OR

A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

(5 Marks)

Q35. The angles of elevation of the top of a tower from two points at a distance of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Prove that the height of the tower is 6 m.

(5 Marks)

SECTION E

Case Study

Q36. A group of students decided to make a project on "Heights and Distances". They used a drone to capture images of a tall building. They noted the following observations:

• The angle of elevation of the top of the building from a point A on the ground is 30°.

• From a point B on the ground, 20 meters closer to the building from point A, the angle of elevation of the top of the building is 60°.

• The points A, B, and the base of the building are in a straight line.

Based on the above information, answer the following questions:

(i) Draw a neat labelled diagram representing the given situation.

(1 Mark)

(ii) If the height of the building is 'h' meters, express the distance of point A from the base of the building in terms of h.

(1 Mark)

(iii) Find the height of the building.

(2 Marks)

OR

(iii) Find the distance of point B from the base of the building.

(2 Marks)

Q37. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 (see figure), and these are equally likely outcomes.

[Imagine a circular spinner with numbers 1 to 8 marked on it, equally spaced.]

Based on the above information, answer the following questions:

(i) What is the probability that the arrow will point at an odd number?

(1 Mark)

(ii) What is the probability that the arrow will point at a number greater than 2?

(1 Mark)

(iii) What is the probability that the arrow will point at a number less than 9?

(2 Marks)

OR

(iii) What is the probability that the arrow will point at a number divisible by 3?

(2 Marks)

Q38. A class teacher has the following absentee record of 40 students of a class for the whole academic year. The number of days a student was absent is given in the table below:

Number of days: 0-6, 6-12, 12-18, 18-24, 24-30, 30-36, 36-42

Number of students: 11, 10, 7, 4, 4, 3, 1

Based on the above information, answer the following questions:

(i) What is the lower limit of the modal class?

(1 Mark)

(ii) What is the frequency of the class succeeding the modal class?

(1 Mark)

(iii) Find the mean number of days a student was absent.

(2 Marks)

OR

(iii) Find the median number of days a student was absent.

(2 Marks)

End of Question Paper